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VELLORE CAMPUS



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Task 5
MPMC LAB

Registration Number: 24BCE0696

Name: Patel Purav Nitinkumar

Experiment Number: 5

Date: 01 April 2026

AIM:

To write, compile and simulate an 8051 microcontroller program using Keil μ Vision software and verify the output using the simulator.

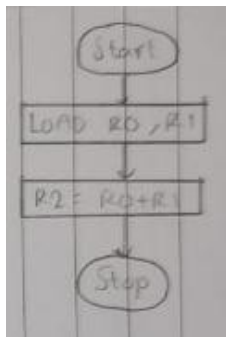
APPARTUS REQUIRED:

Keil μ Vision software

ALGORITHM:

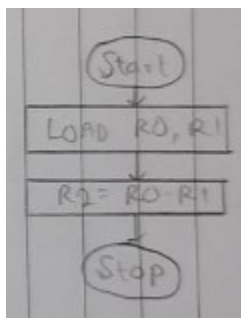
1)

a) Addition



1. Start
2. Load first number into R0
3. Load second number into R1
4. Add R0 and R1 $\rightarrow$ store in R2
5. Stop

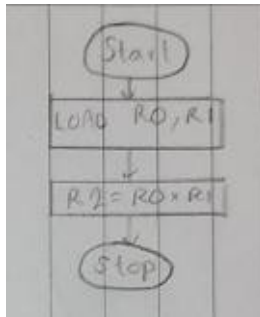
b) Subtraction



1. Start
2. Load R0 and R1
3. Subtract R1 from R0
4. Store result in R2

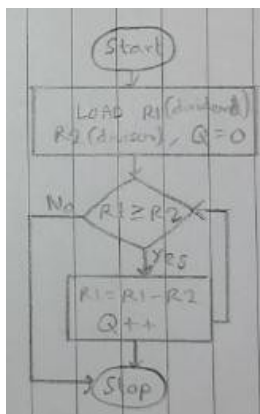
5. Stop

c) Multiplication



1. Start
2. Load operands into R0 and R1
3. Multiply
4. Store result in R2
5. Stop

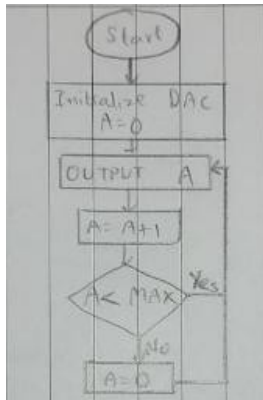
d) Division



1. Start
2. Load dividend in R1
3. Load divisor in R2
4. Initialize quotient = 0
5. Compare R1 and R2
6. If $R1 < R2 \rightarrow$ Stop
7. Else subtract R2 from R1
8. Increment quotient
9. Repeat
10. Stop

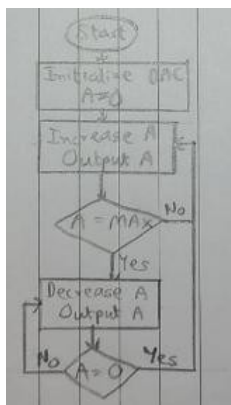
2)

a) Sawtooth



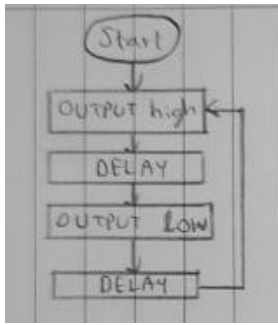
1. Start
2. Initialize DAC
3. Set value = 0
4. Output value
5. Increment value
6. If value < max → repeat
7. Else reset to 0
8. Repeat

b) Triangle



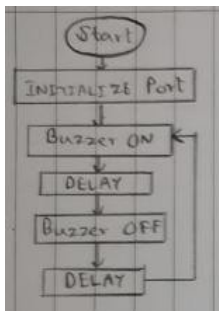
1. Start
2. Initialize DAC
3. Increase value gradually
4. When max reached
5. Decrease value
6. When zero reached
7. Repeat

c) Square



1. Start
2. Initialize DAC
3. Increase value gradually
4. When max reached
5. Decrease value
6. When zero reached
7. Repeat

3)



1. Start
2. Configure port as output
3. Turn buzzer ON
4. Delay
5. Turn buzzer OFF
6. Delay
- 7.** Repeat

CODE:

Q1)

Write ALP and to execute basic arithmetic operations in ARM7 processor

```
i) Write ALP and to execute basic arithmetic operations in ARM7
A: a) Addition-      area addition, code, readonly
                     mov r0, #0x03
                     mov r1, #0x02
                     add r2, r1, r0
                     l b l
                     end

b) Subtraction-      area subtraction, code, readonly
                     mov r0, #0x06
                     mov r1, #0x02
                     sub r2, r0, r1
                     l b l
                     end

c) Multiplication      area multiplication, code, readonly
                     mov r0, #0x03
                     mov r1, #0x02
                     mul r2, r1, r0
                     l b l
                     end

d) Division
                     area division, code, readonly
                     mov r1, #0x05
                     mov r2, #0x02
                     li cmp r1, r2
                     bcc l
                     sub r1, r2
                     add r3, #0x01
                     b eq l
                     l b l
                     end
```

Q 2,3,4) Generation of waveform using ARM processor
(Sawtooth, Triangular, Square)

```
i) Sawtooth wave-      area DAK, CODE, READONLY
                       PINSEL1 EQU 0xE0000006
                       DACR EQU 0xE0000000
                       ldr v3 = 0x000003FF
                       ldr r1 = PINSEL1
                       ldr r2 = 0x00000000
                       str r2, [r1]
                       ldr r0 = DACR
                       ldr r2 = 0x0
                       DtoA
                       ldr r1, r2, #6
                       str r1, [r0]
                       add r2, r2, #1
                       cmp r2, r3
                       blt DtoA
                       ldr r2, =0x0
                       b DtoA
                       end

ii) Triangular wave-      area trianglewave, CODE, READONLY
                          PINSEL1 EQU 0xE0000006
                          DACR EQU 0xE0000000
                          ldr v3 = 0x000003FF
                          ldr r1 = PINSEL1
```

```

ldr r2, #0x00000000
ldr r1, [R1]
ldr r0, #0x00000000
ldr r2, #0x00000000
DtoAp
lsl R1, R2, #6
str R1, [R0]
add R1, R2, #1
cmp R2, R2
BLT DtoAp
DboAn
lsl R1, R2, #6
str R1, [R0]
sub R2, R2, #1
BNE DboAn
B AtoAp
END

```

ii) Square waveform using DAC of an ARM7 processor and verify on processor board -

```

area name, code, readonly
io0dir equ 0x0028008
io0set equ 0x0028004
io0clr equ 0x002800c
io0pin equ 0x0028000

start
ldr r0, #io0dir
ldr r1, #0xffffffff
str r1, [r0]
ldr r2, #io0pin
str r1, [r2]
bl delay
ldr r1, #0x00000000
ldr r3, #io0pin
str r1, [r3]
bl delay
b start

delay
ldr r4, #0x0000ffff
l sub r4, r4, #0x01
bne l
bx lr
done b done
end

```

Q5) To Write an assembly language program to interface a buzzer with an ARM7 processor and make the buzzer on and off with human perceivable delay.

```

CODE: area buzzer, code, readonly
IO1DIR EQU 0x00000018
IO1SET EQU 0x00000014
IO1CLR EQU 0x0000001c
IO1PIN EQU 0x00000010
START
LDR R0, #IO1DIR
LDR R1, #0x00000000
STR R1, [R0]
LDR R2, #IO1SET
STR R1, [R2]

```

```

BL DELAY
LDR R3, #IO1CLR
STR R1, [R3]
BL DELAY
B START
DELAY
LDR R4, #0x0000000f
L SUBS R4, R4, #0x01
BNE L
BX LR
END

```

RESULT:

The programs were successfully executed and analyzed.